

AADIP THAPALIYA

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Aadip

PROFILE

Data Science student specializing in causal inference and time series forecasting. Built an open-source reporting tool for non-technical users, causal estimation pipelines with sensitivity analysis, and forecasting systems across 20 equities.

WORK EXPERIENCE

Machine Learning Research Intern

Varun Herbal Pvt. Ltd.

Jan 2025 – Jul 2025

Remote

- Engineered an automated **Python pipeline** for daily data ingestion and reporting, cutting manual effort by **40%** across all operational records.
- Built a **ML classifier** on **10,000+** samples achieving **90% accuracy** in crop yield risk prediction with zero manual intervention.
- Delivered automated **risk reports** identifying key yield loss drivers, contributing to an estimated **15% reduction** in production losses.

PROJECTS

Causal-Inference: Causal Impact Estimation Pipeline

2026

- Designed a reproducible workflow** to study causal effects of economic policies, from literature review through model benchmarking.
- Developed a causal pipeline** combining IPW, AIPW, and meta-learners to estimate heterogeneous treatment effects.
- Applied Rosenbaum bounds** sensitivity analysis across **five policy datasets** to validate causal claims against unmeasured confounding.

CLUE: Automated Forecasting & Analysis Pipeline

2025

- Designed CLUE**, an open-source tool automating complex data workflows so non-technical users can access insights without coding.
- Built a modular pipeline** for automated cleaning, forecasting, and PDF report generation, eliminating manual spreadsheet dependency.
- Integrated multiple data sources** into intuitive dashboards delivering actionable outputs tailored to end-user decision-making needs.

Financial Time Series Forecasting: Multi Model Benchmark

2025

- Led the full analytical lifecycle** from literature review through model development for financial time series forecasting.
- Benchmarked 10+ forecasting models** including ARIMA, GARCH, XGBoost, LightGBM, LSTM, N-BEATS, and Prophet on stock return data across **20 equities**.
- Achieved best results with LightGBM**, securing **0.034 MAE** and **0.046 RMSE**, outperforming both statistical and deep learning baselines.

EDUCATION

B.Sc. Digital Business & Data Science

University of Europe for Applied Sciences

2024 – 2027

Potsdam, Germany

Machine Learning Study Programme

opencampus.sh (CAU Kiel)

Ongoing

Kiel, Germany

TECHNICAL SKILLS

PythonSQLRExcel

PandasNumPyPostgreSQL

Scikit-learnTensorFlowPyTorch

Time Series ForecastingCausal Inference

Data VisualisationPowerBIAutomation

CORE COMPETENCIES

- Causal Inference & Sensitivity Analysis
- Time Series Forecasting & Benchmarking
- Automated Data Pipeline Development
- End-to-End Analytical Workflows
- Research Reporting & Data Visualisation

CERTIFICATES

Machine Learning with TensorFlow

opencampus.sh

Predictive Modeling, Classification, NLP

Advanced Time Series Forecasting

opencampus.sh

Data Modeling, Forecasting, Reporting

Scientific Machine Learning

opencampus.sh

Analytical Modeling, PyTorch, Python

LANGUAGES

English (C1)

German (B1)